

A Printable Silicon–MXene | Garnet-Composite | NMC Lithium-Ion Cell: An EV-Directed Chemistry

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Abstract

This paper specifies a lithium-ion chemistry aimed at electric-vehicle energy scale, built from the printable MXene architecture developed in earlier drafts. MXene is no longer the energy-storing anode. Silicon or silicon suboxide is the lithium host on the negative side. $\text{Ti}_3\text{C}_2\text{T}_x$ is a 2D conductive, compliant scaffold that wires Si particles and takes part of their volume change. The positive electrode is NMC 811. The separator is a thin, LLZTO-rich PVDF–PEO–LiTFSI film with a ceramic-weighted cathode face. Lithium lost to SEI and first-cycle trapping is supplied by prelithiation of the anode or by a sacrificial source.

The working ion is Li^+ . Design full-cell average voltage is 3.3–3.6 V. A conservative stack estimate at practical loadings sits near 180–240 Wh kg^{-1} . Crossing 250 Wh kg^{-1} at cell level requires higher Si utilization, less foil and polymer, and a thinner ceramic-leaning separator than the first printed prototype. Packaged 400 Wh kg^{-1} and 10C continuous charging are not claimed. The text is a materials specification and a development sequence, not a demonstration.

Keywords: silicon anode; $\text{Ti}_3\text{C}_2\text{T}_x$; NMC 811; LLZTO; composite polymer electrolyte; prelithiation; EV cell

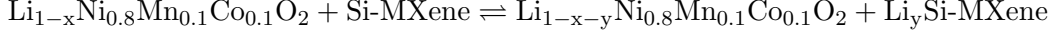
1 Purpose

A printable $\text{Ti}_3\text{C}_2\text{T}_x/\text{Nb}_2\text{CT}_x$ anode facing NMC is scientifically coherent and energy-poor for a long-range pack. MXene stores Li at mid potential. The full cell then sits near 2.5–3.0 V. Electric-vehicle cells that matter today run nearer 3.6 V with graphite or silicon-containing anodes and 200–300 Wh kg^{-1} at the cell.

The move that preserves the printable 2D-scaffold idea and raises voltage is to give the energy-storing job to silicon and keep MXene as infrastructure [1, 2]. That is the chemistry specified here.

2 Cell reaction

On charge, Li leaves NMC and alloys with silicon in the MXene scaffold. On discharge the ions return.



Windows used for design:

- Si/MXene anode: 0.05–1.0 V vs. Li/Li⁺, most capacity below 0.5 V after formation;
- NMC 811: 3.0–4.15 V vs. Li/Li⁺ against a PEO-containing membrane (4.20 V only with a verified ceramic cathode face);
- full-cell averages after matching: 3.3–3.6 V.

3 Negative electrode

3.1 Roles

Silicon (or SiO_x, 0.8 ≤ *x* ≤ 1.1) is the Li host. Theoretical capacity of Si is 3579 mAh g^{−1} to Li₁₅Si₄. Practical reversible capacity in a composite that survives cycling is much lower. This specification uses 800–1200 mAh g^{−1} of Si/SiO_x mass and 500–700 mAh g^{−1} of whole dry anode for matching.

Ti₃C₂T_x does three jobs that silicon cannot do alone [1]: metallic wiring through a swelling electrode, a sheet network that limits electrical isolation of cracked Si, and a 2D surface that can pass a short-range Li flux. It is not counted as the primary capacity.

3.2 Dry composition

Component	wt% of dry electrode
Nano-Si or SiO _x (< 150 nm, C-coated preferred)	55
Delaminated Ti ₃ C ₂ T _x (O-rich)	18
Carbon (nanoribbons or nanotube + carbon black)	12
PVDF or CMC/SBR (aqueous process) / PVDF (NMP)	15

Target loading for an EV-directed prototype: 2.5–4.0 mAh cm^{−2} reversible after formation, which at 600 mAh g^{−1} electrode is about 4–7 mg cm^{−2} dry anode. Porosity 35–45% to leave room for Si swell. Thickness 40–70 μm after calendaring.

SiO_x is the safer first powder (less swell, lower capacity, easier SEI). Nano-Si is the energy upgrade once the scaffold and prelithiation work.

3.3 Why this is buildable

Si/MXene composites are an established experimental family: MXene wraps or interleaves Si, raises electronic conductivity, and improves cycling relative to bare nano-Si [1, 2]. Half-cell

numbers in that literature are often high at low loading. This paper uses conservative full-electrode capacities and insists on areal loading that an EV cell would actually need.

3.4 Ink and process

Aqueous CMC/SBR with MXene is compatible with MXene colloids if the pH is controlled and oxygen is limited. NMP/PVDF is the drier-room alternative. Short ice-bath dispersion, no long sonication that fractures Si and oxidizes MXene. Dry under inert gas 80–120 °C. Calender to the target porosity; over-calendering closes pores that Si needs.

4 Positive electrode

4.1 Dry composition

Component	wt% of dry electrode
NMC 811, single-crystal or large-grain preferred	92
Carbon black	4
Ti ₃ C ₂ T _x (optional wiring)	0–2
PVDF	4

Optional particle coating: LiNbO₃, Li₃PO₄, or a thin LLZTO sol on the NMC to cut PEO oxidation and Ni-rich surface attack.

Target: 3.0–4.5 mAh cm⁻² at 160–180 mAh g⁻¹ of oxide in the solid cell (4.15 V cap). That is 18–28 mg cm⁻² oxide—standard EV cathode loading, not a thin printed demo.

LMFP is an allowed substitute if cobalt-free chemistry is required. Voltage and energy fall slightly; oxidative stress on PEO falls with them.

5 Separator

5.1 Baseline (printable, EV-directed)

Component	wt% of dry membrane
PVDF + PEO (75:25)	50
LiTFSI, EO:Li = 16	set by PEO mass
Silane-modified LLZTO	35
Ti ₃ C ₂ T _x	2
Cross-linker residue	~1

Thickness 15–25 μm. Ceramic-rich skin on the NMC face. Electronic conductivity ≤ 10⁻⁹ S cm⁻¹. Room-temperature ionic conductivity specified as ≥ 3 × 10⁻⁴ S cm⁻¹ for a prototype; 10⁻³ is the EV target, not the first measurement.

PEO remains the printable continuum. At 35 wt% LLZTO the ceramic is intended to percolate. If it does not, the film is only a filled polymer and EV rate will fail.

5.2 Next separator (EV cell, not the first print)

A tape-cast or aerosol-deposited LLZTO or halide film, 10–20 μm , with a thin polymer wetting layer on each electrode. Sulfide/argyrodite is chemically stronger as a conductor and hostile to ambient printing. It is a factory-atmosphere branch, not the baseline of this paper.

6 Lithium inventory

Silicon SEI and the first-cycle irreversible capacity of SiO_x will consume a large fraction of the Li in NMC if the anode is assembled empty. Required before EV-relevant cycle life:

1. electrochemical or chemical prelithiation of the Si/MXene film to a defined SOC, sized to measured first-cycle loss plus $\geq 10\%$ reserve; or
2. a sacrificial Li source in the cathode, counted in the mass budget; or
3. formation in a temporary Li-metal half-cell, then reassembly—laboratory only.

Prelithiation is part of the chemistry, not an optional extra.

7 Current collectors and format

Positive: 12 μm Al. Negative: 8–10 μm Cu (required once the anode sits below ~ 0.5 V vs. Li; Al is a risk). Pouch, stacked, with modest uniaxial pressure (0.1–1 MPa class) to keep Si in contact with the MXene network. Bipolar packaging is not assumed.

8 Conservative energy estimate

Illustrative EV-directed stack (one layer, no pouch foil):

Layer	μm	mg cm^{-2}	Role
Al foil	12	3.2	pos. collector
NMC electrode	70	24	4 V host
Separator	20	3.0	LLZTO-rich CSE
Si/MXene electrode	55	6.0	Si host + scaffold
Cu foil	8	7.2	neg. collector
Stack	165	43.4	—

If the anode delivers 600 mAh g^{-1} of 6.0 mg cm^{-2} , areal capacity is 3.6 mAh cm^{-2} . Matched NMC at 165 mAh g^{-1} of 22 mg cm^{-2} oxide in the 24 mg layer is 3.6 mAh cm^{-2} . At 3.45 V average, areal energy is 12.4 mWh cm^{-2} . Divided by 43.4 mg cm^{-2} this is ~ 286 Wh kg^{-1} *if* those capacities and that voltage are all realized in the solid printed cell.

That “if” is the paper. In liquid electrolyte, Si/MXene and NMC can approach those numbers at these loadings in careful cells. In a 20 μm polymer–garnet film with Si swell, first-cycle loss, and a 4.15 V cap, a first honest cell-level band is **180–240** Wh kg^{-1} after formation.

Pack level after tabs, pouch, cooling, and module structure is lower. This is the EV conversation range only after the solid interfaces work. It is not a CES slide.

Rate: design continuous 1C, pulse 2–3C, after the separator measures $\geq 3 \times 10^{-4} \text{ S cm}^{-1}$ and Si loading is not pushed above the scaffold’s mechanical limit. 10C at this energy is not specified.

Cycle life: Si swell is the limiter. Target for a development cell is 80% retention at 500–1000 cycles at 80% DoD, not 10^5 cycles.

9 Development sequence

1. Liquid-electrolyte half-cells of the printed Si/MXene anode at 3 mAh cm^{-2} . Measure reversible capacity, first-cycle loss, and thickness change.
2. Same anode vs. NMC in liquid electrolyte. Full-cell voltage and inventory.
3. Standalone LLZTO-rich membrane: σ_{ion} , σ_e , LSV to 4.3 V.
4. Laminated solid full cell at low loading, then at 3 mAh cm^{-2} .
5. Only then: thinner ceramic film, higher Si fraction, pouch stack, thermal and nail tests.

A lithium-metal anode with a MXene interlayer is a later branch for $> 250 \text{ Wh kg}^{-1}$. It is not the first EV chemistry in this paper.

10 What this chemistry is not

It is not a MXene-only anode cell. It is not a proton or Grotthuss cell. It is not V_2O_5 -cathode chemistry. It is not a demonstration of 400 Wh kg^{-1} , 100 000 cycles, or lithium-free operation. Those claims do not follow from Si + MXene + NMC + LLZTO.

11 Conclusion

The EV-directed descendant of the printable MXene hybrid is a conventional lithium-ion couple with an unconventional anode architecture: silicon stores the lithium, MXene holds the electrode together and conducts, NMC sets the high potential, and a ceramic-rich printable film separates them. Voltage and capacity can, in principle, enter the band where a traction pack is discussable. Whether they do depends on prelithiation, Si swell in a solid stack, and whether PEO can be kept off a 4.2 V NMC surface. Those are the experiments. The chemistry is specified so they can be run without moving the energy-storing role back onto MXene.

References

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